ANALYTICA CHIMICA ACTA 629 (2008) 119–127

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Comparison of near-infrared and Raman spectroscopy for on-line monitoring of etchant solutions directly through a Teflon tube

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ARTICLE INFO

Article history:
Received 2 August 2008
Received in revised form
4 September 2008
Accepted 10 September 2008
Published on line 18 September 2008

Keywords:
Etchant solution
Near-infrared spectroscopy
Raman spectroscopy
Prediction robustness
Moving window partial least squares
Spectral selectivity

ABSTRACT

Both near-infrared (NIR) and Raman spectroscopy have been studied for the quantitative measurement of components (H_3PO_4, HNO_3, and CH_3COOH) in an etchant solution and the corresponding prediction robustness has been evaluated. Both measurements were accomplished by illuminating radiation directly through a Teflon tube. Raman spectral features of each component were much clearer and more selective than those observed in the NIR spectrum. Especially, NIR spectral variation pertinent to H_3PO_4 and HNO_3 were mostly based on the displacement and perturbation of water bands rather than due solely to NIR absorption. Therefore, the resulting spectral variations were not highly specific. When the validation set contained minor spectral variations resulting from a slight instrumental change, NIR prediction performance for all three components degraded substantially by showing obvious prediction bias. However, the accuracies of Raman predictions were maintained. Since partial least squares (PLS) models for each component were built using NIR spectra of poor specificity with broadly overlapping features, even minor spectral differences introduced by instrumental variations sensitively influenced the prediction performance of the NIR models. Overall, the selectivity (specificity) of a targeting spectroscopic method should be considered critically to secure prediction robustness for monitoring components in an etchant solution.

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1. Introduction

Near-infrared (NIR) spectroscopy has recently been studied for on-line monitoring of components in etchant solutions; these solutions are used extensively in semi-conductor industries to remove impurities in silicon wafers. These etchants are mostly oxidants and diverse acids in an aqueous medium. By providing direct NIR radiation through an existing Teflon line, measurements can be taken without any process modifications, as shown in previous studies [1,2]. In general, the NIR spectral features of the components in etchant solutions broadly overlapped with one other and the water spectral feature. Sometimes the perturbation and/or displacement of the water band was the only information available to build a model for some components, such as H_3PO_4 and HNO_3, which had no separate NIR absorption. Under this circumstance, the selectivity of a multivariate calibration model should be careful considered. Based on a previous study, the spectral range should be carefully determined to achieve an acceptable selectivity [3,4]. When a calibration model is built using spectra of

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doi:10.1016/j.aca.2008.09.032

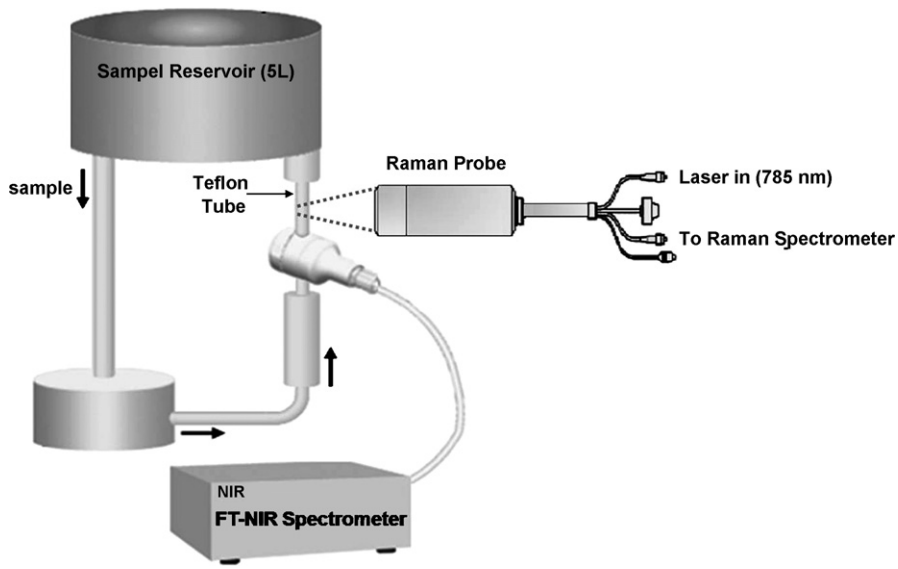

Fig. 1 – The graphical description of a chemical wet bench for circulating samples and the spectral collection scheme for acquiring NIR and Raman spectra directly from a Teflon tube.

low selectivity and sensitivity, such as NIR spectra of etchant solutions, it is highly likely that prediction performance would be affected by minor spectral variations other than the compositional change in the components. Therefore, the long-term robustness of NIR measurements for monitoring etchant solutions is another important issue to consider.

Alternatively, Raman spectroscopy can provide more distinct spectral features (better selectivity) than NIR spectroscopy, especially for inorganic acids, as shown previously [5]. To exploit the superior selectivity of Raman spectroscopy, a Raman collection scheme capable of providing reproducible Raman spectra should be developed. Direct Raman spectral collection through a Teflon tube is advantageous because a non-overlapping Teflon band could be used as a standard to correct Raman intensity variation which is essential for reliable quantitative analysis [6].

In this publication, the analytical performance of NIR and Raman spectroscopy for determining the compositional analysis of a metal aluminum etchant (MAE, a mixture of H_3PO_4, HNO_3, and CH_3COOH) solution, one of the most frequently used etchant solutions, was compared. NIR and Raman spectra of MAE solutions were collected synchronously in an on-line manner by circulating samples using a pilot scale chemical wet bench, as shown in Fig. 1. For NIR measurements, NIR radiation from an optical fiber was illuminated in the center of the Teflon tube. The transmitted radiation was also collected with the use of an optical fiber. The sample holder was designed to hold the Teflon tube firmly. To maintain reproducibility of the Raman measurement, the wide area illumination (WAI) Raman scheme that applied a laser in a circular fashion with a 6-mm diameter (area: 28.7 mm^2) with a focal length of 250 mm, was employed. Previous studies have shown that the use of the WAI scheme helps to improve the reproducibility and accuracy of the Raman measurement [6–9]. As shown in Fig. 1, NIR and Raman spectra were collected simultaneously through the same Teflon tube to ensure fair comparison of the two methods for quantitative analysis of the components in the MAE solution. Using both NIR and Raman spectra, partial least squares (PLS) models [10,11] were built and the corresponding models were validated by predicting the spectra collected after 2 months of initial spectral collection, where some minor spectral variations were observed.

2. Experimental

2.1. *NIR and Raman spectral collection*

Fig. 1 shows the schematic diagram of the pilot chemical bench used for this study. This chemical bench was designed to control the temperature and flow rate while continuously circulating an etchant sample through a 0.75-in. Teflon tube. The volume of the reservoir (also made of Teflon) was 5 L. Both NIR and Raman spectral collections were performed in the same Teflon tube with continuous flow (flow rate: 4 L min^{-1}) of the sample.

For the NIR measurements, a fiber optic probe designed to transmit NIR radiation through the Teflon tube was used. The tube was held firmly in place by the probe. A detailed description of the fiber optic configuration can be found in a previous publication [1]. An ABB Bomem fiber optic-based FT-NIR spectrometer equipped with a tungsten–halogen source and a Germanium detector was used. All NIR spectra corresponded to an accumulation of 128 scans with a resolution of 32 cm^{-1} (collection time: 60 s). Five spectra were collected per sample.

For Raman spectral collection, the WAI scheme (Kaiser Optical, Inc., Ann Arbor, MI) was used in which a laser beam is used to illuminate a sample in a circular fashion, with a 6-mm diameter (area: 28.3 mm^2) and a focal length of 250 mm. A Teflon tube containing the circulating sample was located in

the center of the focal point, as shown in Fig. 1. Raman spectra with a resolution of 4 cm^{-1} were collected using a 785-nm laser with an exposure time of 7 s and nine scans (collection time: 63 s). Five spectra were collected per sample.

2.2. *Sample preparation*

For the preparation of the calibration set, 11 MAE samples were prepared by mixing appropriate amounts of H_3PO_4, HNO_3, and CH_3COOH in water. The concentration of each component was determined by generating 11 random numbers in a given concentration range. The concentrations of H_3PO_4, HNO_3, and CH_3COOH ranged from 69.7 to 49.7 wt.%, 8.0 to 1.2 wt.%, and 10.0 to 2.0 wt.%, respectively. When a multivariate calibration model is developed from standard mixtures, it is crucial to have random concentration distributions with no correlation among components. The degree of correlation between the components was determined by performing linear regression analysis with data produced by plotting the concentration of one component versus that of the other two components. In all of the cases, the R^2 values were less than 0.02 and no significant inter-correlations were observed.

To evaluate the robustness of prediction, nine new samples and the corresponding NIR and Raman spectra were collected as described in Fig. 1. The spectral collection was completed 2 months after the initial spectral collection for the calibration set. The NIR fiber optic probe and WAI Raman probe were reinstalled for the spectral collection. Considerable effort was made to maintain the original position of these two probes. The newly collected spectra (45 spectra from nine samples) were used as the validation set. All calculations, including the baseline correction, intensity normalization, and PLS were accomplished using Matlab, version 7.0 (The MathWorks, Inc., Natick, MA).

3. Results and discussion

3.1. *NIR and Raman spectral features of pure components*

Fig. 2 shows the NIR spectra (1100–7500 cm^{-1}) of 30% phosphoric acid (H_3PO_4), 30% nitric acid (HNO_3), 30% acetic acid (CH_3COOH) and water (H_2O). As found in a previous study, two broad water bands were clearly observed at 10,300 and 8400 cm^{-1}, corresponding to the second overtone OH band and the combination OH band, respectively [4]. By comparing these features to the water spectral feature, the major source of spectral variations originating from H_3PO_4 and HNO_3 were mostly based on the perturbation and/or displacement of water bands rather than absorption. Therefore, the corresponding spectral features were not distinctly different. In the spectra of H_3PO_4 and HNO_3, the absorption of two water bands decreased due to water displacement. Although no distinct peaks were available for H_3PO_4 and HNO_3, the features of perturbed water bands were slightly different, especially in the 8700–7700 cm^{-1} range. These differences enabled the quantification of H_3PO_4 and HNO_3 concentrations without their own absorption features. The spectral variation according to CH_3COOH was relatively more characteristic due to the

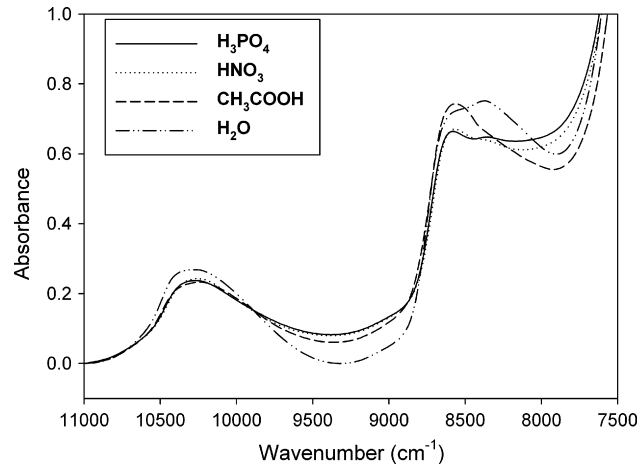

Fig. 2 – NIR spectra (1100–7500 cm^{-1}) of 30% H_3PO_4, 30% HNO_3, 30% CH_3COOH and H_2O.

C–H groups. The C–H band at 8600 cm^{-1} was clearly observed and the decrease of water bands by water displacement was observed simultaneously. The spectral variation of CH_3COOH was relatively more specific than H_3PO_4 and HNO_3.

Fig. 3 shows Raman spectra of 30% H_3PO_4, 30% HNO_3, and 30% CH_3COOH. A water spectrum was not shown as it had no significant Raman features (very weak Raman scatter). These spectra were collected using a quartz cuvet to examine pure features without the superimposition of Teflon bands. For comparison, the Raman spectrum of Teflon tube are also displayed. The spectral features of H_3PO_4 and HNO_3 are clearly present with major bands at 898 and 1047 cm^{-1}, corresponding to the stretching mode of P=O and symmetric stretching of the nitrate ion, respectively. Since their molecular structures are clearly different and the corresponding vibrations are Raman-active, separate Raman bands with reasonable intensity can be observed. There are two major bands for CH_3COOH at 891 and 625 cm^{-1}, which correspond to stretching of the C–C backbone and rocking of the carboxylic group, respectively. The CH_3COOH band at 891 cm^{-1} overlaps closely with H_3PO_4 at 898 cm^{-1}.

By comparing the NIR spectral features of each component, the corresponding Raman spectral features are much more selective and less overlapping. Therefore, the selectivity and robustness of the Raman measurements can be enhanced as long as reproducible Raman spectra can be collected with a proper scheme to correct the variation of Raman intensities originating from the laser power change.

The notable advantage of the Raman collection scheme in Fig. 1 is the use of a Teflon band to correct the Raman intensity variation that results from occasional laser power changes. If at least one non-overlapping Teflon peak is available, the peak can be used to compensate for variation in Raman intensity in a synchronous fashion. The dotted vertical lines on the Raman spectra of three pure components correspond to the peak positions of Teflon bands. Since the Teflon peak centered at 290 cm^{-1} did not overlap seriously with the bands of the three components, it was used as the standard peak to correct Raman intensity variation for further study.

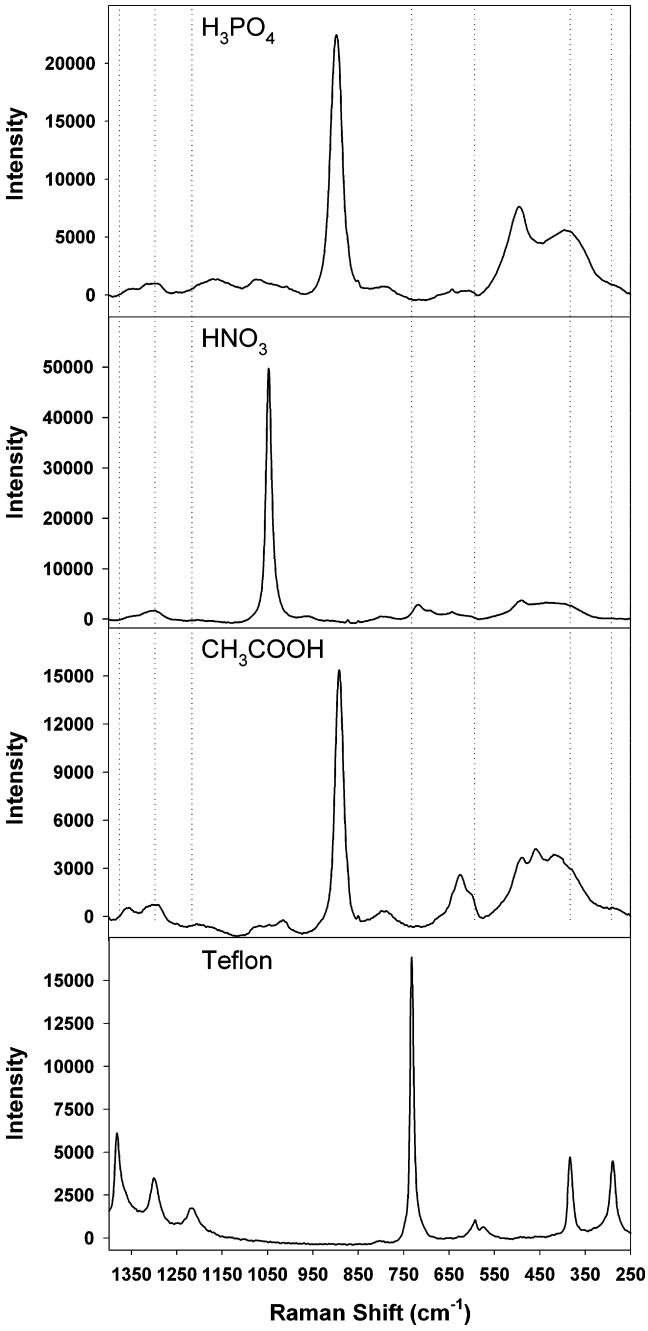

Fig. 3 – Raman spectra of 30% H_3PO_4, 30% HNO_3 and 30% CH_3COOH collected using a quartz cell. Raman spectrum of Teflon tube is also displayed. The dotted vertical lines on the Raman spectra of three pure components correspond to the peak positions of Teflon bands.

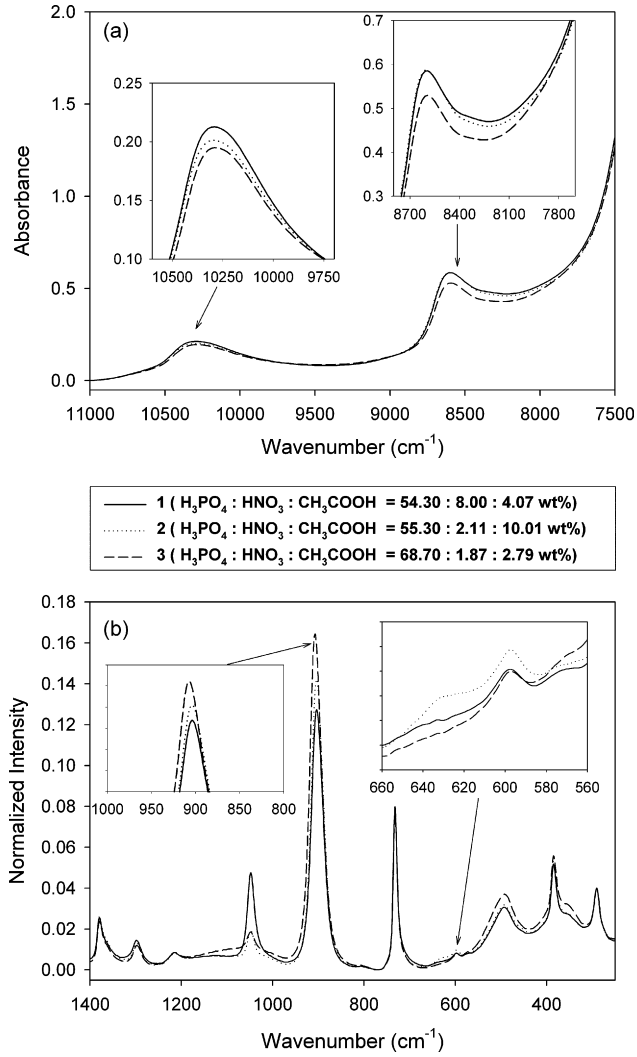

Fig. 4 – (a) NIR and (b) Raman spectra of three selected mixture samples of (a). The corresponding concentrations of three samples are shown.

3.2. *NIR and Raman spectral features of mixtures*

Fig. 4(a) shows NIR spectra of three selected mixture samples with baselines of NIR spectra corrected at 11,000 cm^{-1}. The three samples were selected to show the highest concentrations of each component and the corresponding concentrations are shown in the figure. Two regions of water bands are enlarged for detailed examination. Similar to Fig. 2, the major variation in the NIR spectra is primarily from the displacement of water bands depending on the combined amount of the acids. The two water bands of Sample 3 (the highest amount of acid) show the lowest water absorption due to the highest degree of water displacement. The degree of water band displacement near 8400 cm^{-1} for Samples 1 and 2 were similar as the amount of acid was similar; however, the band shapes were slightly different. This difference originated from the slightly different perturbation of water band by each component. Consequently, the main source of information to quantify each component using NIR spectra was the displacement and perturbation of the water bands, which is not highly selective, as shown in the figure.

Fig. 4(b) shows normalized Raman spectra of the same three mixture samples used for examination of NIR spectral features. The Raman spectra were a result of the summation of spectral features of each component and the Teflon tube. Initially, baselines of raw Raman spectra were corrected at 770 cm^{-1} and the area of the Teflon band between 320

and 270 cm^{-1} was calculated. Each sample spectrum was then divided by the corresponding area of the Teflon band, which effectively corrected for unwanted intensity variations from the laser power change. The regions of 1000–800 cm^{-1} and 660–560 cm^{-1} were enlarged for detailed examination of the spectral variation of HNO_3 and CH_3COOH, respectively.

The HNO_3 band at 1047 cm^{-1} clearly changed according to the concentration variation (highest intensity for Sample 1). Sample 3, containing the highest concentration of H_3PO_4, had the most intense band near 900 cm^{-1} that overlaps with the band of CH_3COOH at 891 cm^{-1}. The position of the band shifts to a lower wavenumber by overlapping with the band of CH_3COOH. Another selective band for CH_3COOH was located at 625 cm^{-1}, although the intensity was low. Sample 2 which contains the highest CH_3COOH concentration clearly shows the highest intensity band at 625 cm^{-1}. Additionally, the intensities of Teflon at 290 cm^{-1} were nearly the same due to normalization. Overall, the Raman intensity changes according to their concentrations with clear spectral features.

The comparison of NIR and Raman spectral features of mixtures samples indicates that the Raman spectral features of the components were distinct, although major bands of H_3PO_4 and CH_3COOH were closely overlapping; while, it is difficult to identify the spectral features of individual components in the NIR mixture spectra. When a multivariate model is constructed using character-deficient spectra such as NIR spectra of MAE solution, the prediction performance is expected to be sensitive to spectral variations other than the concentration variation, such as a slight change in instrumental configuration as well as temperature.

3.3. *Characterization of the spectral differences between the calibration and validation sets*

The possible spectral differences between the calibration and validation set collected on separate periods could be existed. For this purpose, two samples with the same concentrations in the calibration and validation set were selected (referred to as Samples A and B). Fig. 5(top) shows NIR spectra of these two samples. Baselines of spectra were corrected to 11,000 cm^{-1}. Only two spectral regions where the most spectral information was located are highlighted (10,600–9800 cm^{-1} and 8750–8000 cm^{-1} range). The solid and dashed line correspond to the spectra in the calibration and validation set, respectively. The spectral features of Sample A in the calibration and validation set are nearly similar; while the absorbance of Sample B in the calibration set was slightly higher. Seven more samples were used to evaluate this difference more extensively. Generally, the absorption of samples in the calibration set was slightly greater and the difference was consistent for all seven samples.

As shown in Fig. 5, some changes in the condition of the instrumentation between the two spectral collections were observed. Because the NIR measurement was accomplished by transmitting NIR radiation through a round Teflon tube, the actual pathlength could be altered when the horizontal position of the Teflon tube with respect to beam path was slightly changed. This could induce a slight change in the magnitude of the absorption. The same instrumental configuration between two spectral collections was attempted; however, the resulting minute spectral difference may be unavoidable in a practical situation.

To minimize the spectral difference, the baseline of the NIR spectrum was corrected at 11,000 cm^{-1} and the corresponding spectrum was divided by the area integrated from the 11,000–7500 cm^{-1} range. Using this normalization procedure, the possible spectral variation originating from different pathlengths could be reduced. The resulting normalized NIR spectra of Samples A and B are also shown in Fig. 5(bottom). Although there are very minute spectral differences in the normalized spectra, the spectral features of the same sample are nearly identical and visually similar.

Normalized Raman spectra of Samples A and B are shown in Fig. 6. The solid and dashed line correspond to the spectra in the calibration and validation set, respectively. As described previously, the baseline of the raw Raman spectra was initially corrected at 770 cm^{-1} and the corresponding spectrum was divided by the area of the Teflon band in the 320–270 cm^{-1} range. For a more detailed comparison, two spectral ranges of 1100–1000 cm^{-1} and 600–400 cm^{-1} are shown.

Even after normalization by the area of the Teflon peak, slight spectral differences remained between the spectra in the calibration and validation set. Because Raman spectra are also collected by illuminating the laser beam directly through a round Teflon tube, the illumination (cross-section) volume ratio between a sample and the Teflon tube can vary when the horizontal position of the tube slightly changes. Because the isolated Teflon band is used to normalize Raman intensity variation, the normalized band intensity of a given analyte can be affected when the intensity of the Teflon peak varies due to the positional variation rather than the laser power change. This would account for the Raman spectral difference observed in Fig. 6.

Overall, minute spectral differences were observed when comparing the normalized NIR and Raman spectra of the calibration and validation set. These differences were partially due to the measurement scheme used where spectral collection was performed directly through a round Teflon tube. Practically, some minor spectral variations between separately collected datasets would be encountered. Therefore, the robustness of the measurement is one of the most important issues for determining the reliability of a spectroscopic method, especially when multivariate calibration methods are used to resolve broadly overlapping spectral features.

3.4. *PLS calibration*

A PLS regression was used to build calibration models for each component using both NIR and Raman spectra. Full spectral ranges of NIR (11,000–7500 cm^{-1} range) and Raman (1400–250 cm^{-1} range) data were used to build the PLS models. Normalized NIR and Raman spectra were used as described in Figs. 5 and 6, respectively. PLS results using each full spectral range for both methods are summarized in Table 1 in which the numbers in parentheses correspond to the number of factors used. R^2 values were calculated using the prediction results of each component. The resulting concentration correlation plots for each component using NIR and

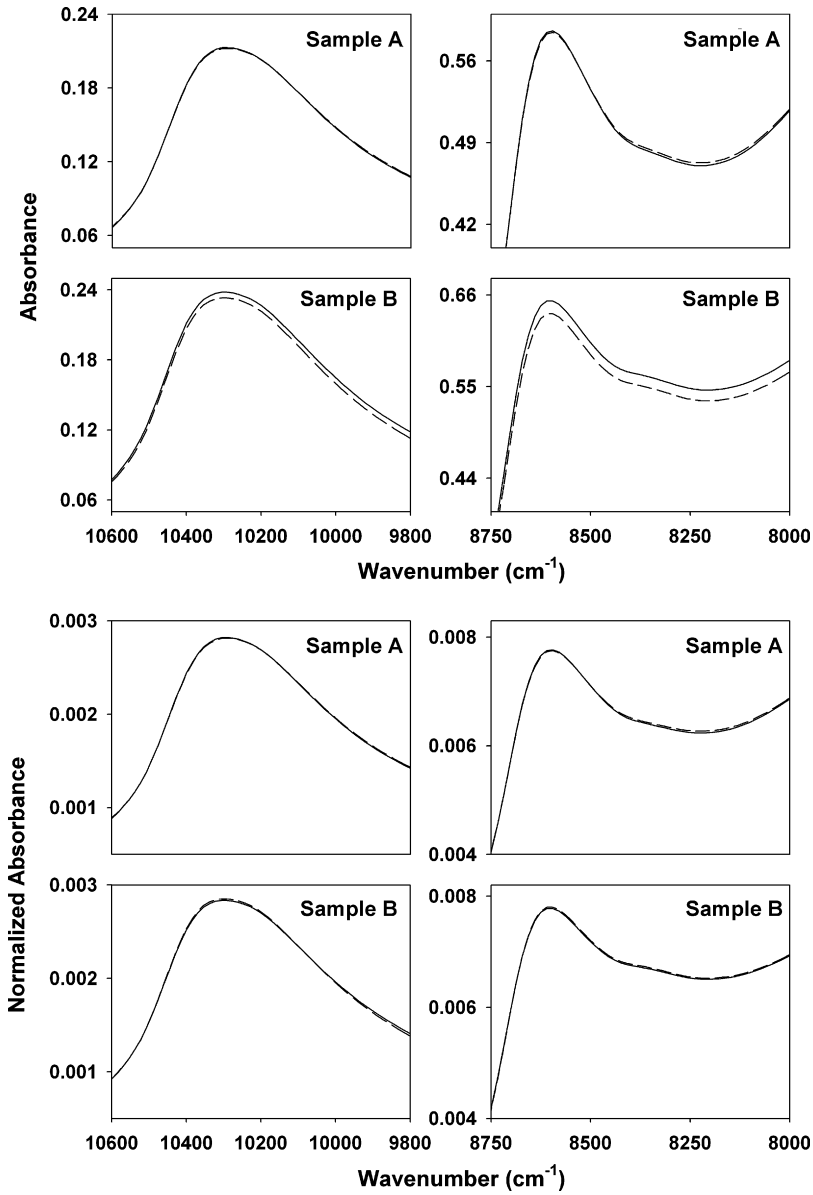

Fig. 5 – NIR spectra (10,600–9800 cm^{-1} and 8750–8000 cm^{-1}) of two selected samples (top) and the corresponding normalized spectra (bottom). The solid and dashed line correspond to the spectra in the calibration and validation set, respectively.

Raman spectroscopy are shown in Fig. 7. Filled and open circles correspond to calibration and prediction data, respectively.

The resulting standard error of calibrations (SECs) from both methods were generally similar, except for the SEC of HNO_3 using Raman spectroscopy. Since a highly selective band at 1047 cm^{-1} is available in the Raman spectrum, the resulting calibration accuracy is superior. When the spectra of the validation set collected on a separate period was predicted, the prediction results from both methods are substantially different. Using NIR spectroscopy, the resulting standard error of predictions (SEPs) were largely increased by presenting obvious bias patterns for all three components, as shown in Fig. 7, although, the spectral shapes of the two datasets were close to each other after normalization. Since PLS models for each component were built using NIR spectra of poor specificity as shown in Fig. 3, even tiny spectral differences between the two datasets sensitively influenced the prediction performance of models by introducing the prediction bias. If the NIR feature of the MAE solution is highly selective with a narrow band width, the prediction accuracy would not be sensitively degraded by the spectral differences present in Fig. 5(bottom). The NIR spectral feature of H_3PO_4 and HNO_3 were quite similar, as shown in Fig. 2, and the degrees of prediction degradation were analogous. While, prediction of CH_3COOH concentrations was less degraded compared to H_3PO_4 and HNO_3. The relatively more selective band of CH_3COOH at 8600 cm^{-1} makes the PLS model less vulnerable to minor and unwanted spectral variation.

Using Raman spectroscopy, the prediction performances were successfully maintained for HNO_3 and CH_3COOH without any bias patterns. Specifically, the prediction of HNO_3 was

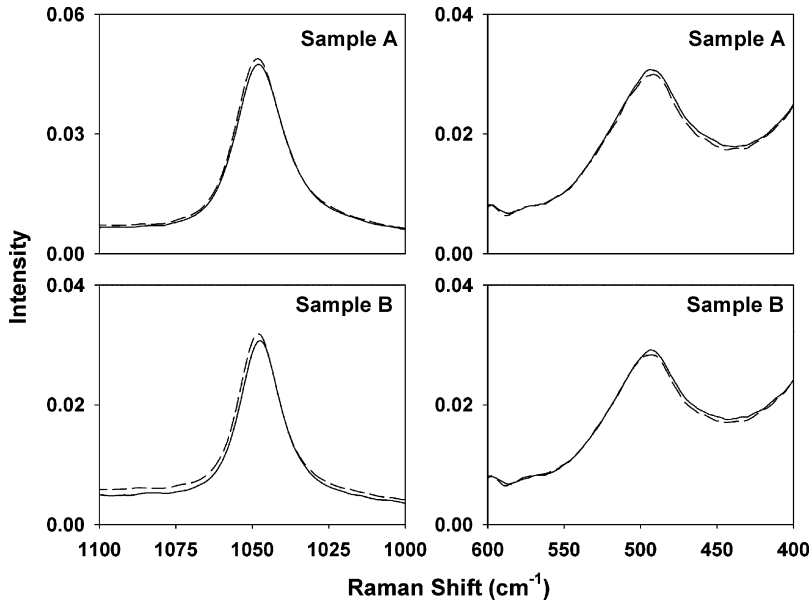

Fig. 6 – Normalized Raman spectra (1100–1000 cm^{-1} and 600–400 cm^{-1} range) of two selected samples. The solid and dashed line correspond to the spectra in the calibration and validation set, respectively.

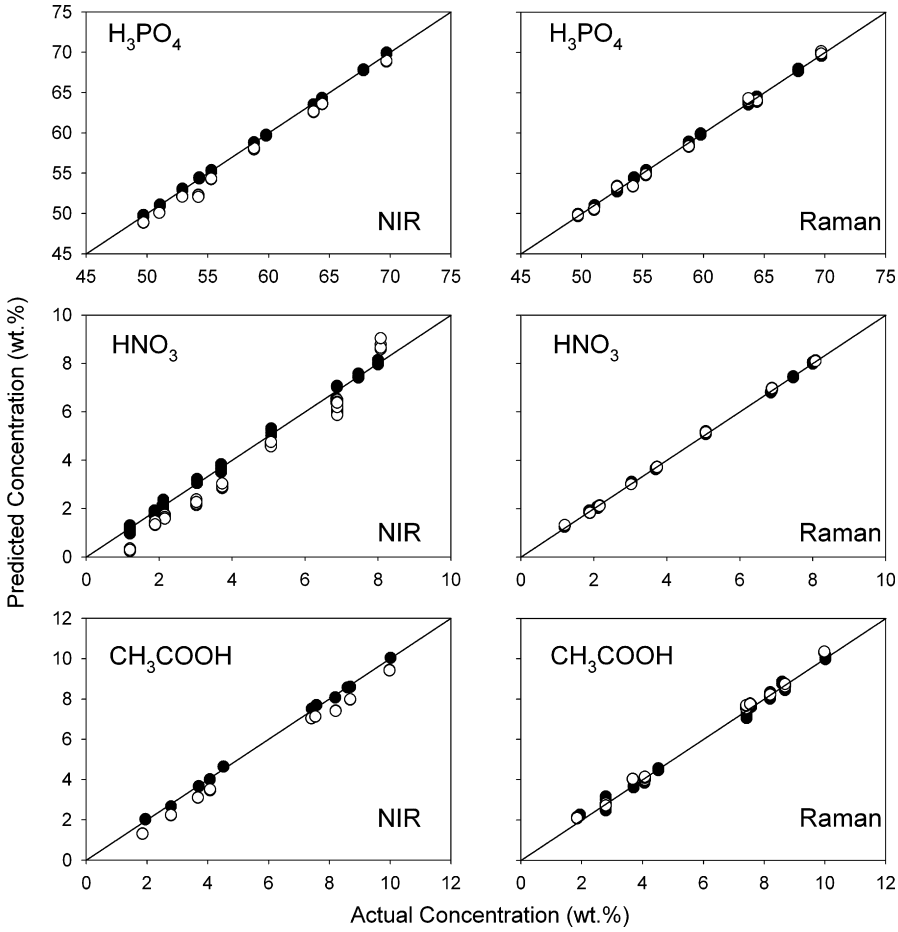

Fig. 7 – Concentration correlation plots for each component using NIR and Raman spectroscopy. Filled and open circles correspond to calibration and prediction data, respectively.

Table 1 – PLS results using full and optimal spectral ranges for both spectroscopic methods

Component	Range (cm^{-1})	SEC (wt.%)	SEP (wt.%)	R^2
NIR				
H_3PO_4	11000–7500	0.15 (7)	0.74	0.9940
	9700–7500	0.08 (7)	1.27	0.9892
HNO_3	11000–7500	0.16 (7)	0.72	0.9756
	9900–7450	0.07 (8)	1.17	0.9534
CH_3COOH	11000–7500	0.11 (7)	0.36	0.9897
	9200–8150	0.10 (8)	0.27	0.9965
Raman				
H_3PO_4	1400–250	0.11 (6)	0.46	0.9958
	1320–270	0.10 (6)	0.33	0.9980
HNO_3	1400–250	0.04(7)	0.07	0.9995
	1360–1030	0.04 (7)	0.11	0.9989
CH_3COOH	1400–250	0.19(5)	0.22	0.9973
	720–570	0.11 (6)	0.47	0.9970

Numbers in parentheses correspond to the number of factors used.

very accurate since the highly specific HNO_3 band at 1047 cm^{-1} can be utilized. The SEP for H_3PO_4 was degraded compared to those of other components. Based on the examination of the concentration correlation plot of H_3PO_4, the increase of SEP was the result of more scattered predictions rather than prediction bias. It is presumed that a selective H_3PO_4 band for robust calibration is relatively deficient compared to other components. As shown in Fig. 3, the most intense H_3PO_4 band at 898 cm^{-1} closely overlapped the CH_3COOH band at 891 cm^{-1} and other lower intensity H_3PO_4 bands also broadly overlapped those of HNO_3 and CH_3COOH. Therefore, the prediction of H_3PO_4 could be more sensitively influenced by minor spectral variations introduced by two separate spectral collections. However, the prediction of H_3PO_4 concentrations was still quite acceptable and the predicted values are close to the ideal line comparing the result of H_3PO_4 using NIR spectroscopy. Although many of the H_3PO_4 bands overlapped, the more selective Raman feature with narrower bandwidth leads to a reasonable prediction performance compared to NIR spectroscopy. For CH_3COOH prediction, the most intense band at 891 cm^{-1} also overlapped with the H_3PO_4 band at 898 cm^{-1} and similarly, the band at 625 cm^{-1} could provide selective information to maintain prediction accuracy.

As observed in the previous NIR study for the MAE solution, the proper selection of a spectral range for PLS substantially improved the selectivity and accuracy of prediction. Therefore, an optimal range for each component using moving window PLS (MW-PLS) was examined in exactly the same way as the previous study [4]. To find an optimal range by MW-PLS, each original calibration set was further divided into a new calibration set (35 spectra of seven samples) and a new validation set (20 spectra of four samples). For each spectral range, a PLS was performed using the new calibration set and the resulting SEP from the new validation set was evaluated to determine the optimal spectral range. PLS was again performed using the original calibration set within the optimal spectral range and the final SEP was calculated by predicting samples in the validation set. Similarly, the optimal ranges of each component were determined for Raman spectral data. For the NIR data, the window size (number of spectral points) varied from 200 to 3600 cm^{-1} in 50 cm^{-1} increments. For the Raman data, the window size varied from 50 to 1150 cm^{-1} in 20 cm^{-1} increments. PLS results using the found optimal ranges are also summarized in Table 1.

When the optimized spectral ranges were used for PLS, the resulting calibration errors were generally improved or similar; however, the resulting prediction results varied. For H_3PO_4 and HNO_3 using NIR spectroscopy, the resulting SEPs further degraded with the larger prediction biases. The optimization procedure only maximizes the correlation between the feature-deficient spectral features and the corresponding concentrations in the calibration set. Therefore, the PLS models would easily overfit and be highly sensitive to other spectral variations present in the validation set. The optimized range (9200–8150 cm^{-1}) for CH_3COOH includes only the available selective band at 8600 cm^{-1} and the corresponding SEP improved using this range. In contrast to the cases of H_3PO_4 and HNO_3, the use of the selective CH_3COOH band helps to improve the robustness of the calibration.

The Raman spectral optimization for H_3PO_4 also improves the resulting SEP. The optimized range (1320–270 cm^{-1}) was close to the full spectral range. For HNO_3 and CH_3COOH, the optimized ranges were well matched to the regions where selective bands were individually located, however, the resulting SEPs did not improved. If the spectral features in the calibration and validation set were purely from the compositional variation without superimposition of instrumental or other variations, the use of an optimized range should provide more accurate calibration results. Since spectral variations other than compositional variation were clearly present in the validation set, as shown in Fig. 6, the use of a full range was relatively more efficient to leverage the unavoidable spectral variations compared to the use of narrow range of a concentrated analyte band. Spectral variation unrelated to the compositional variation could be more amplified when a narrow range is used for PLS.

The spectral variations shown in Figs. 5 and 6 could be minimized by carefully maintaining the instrumental configuration, however, the collection of variation-free NIR or Raman spectra would not be realistic in practical routine and on-line analysis. As stated, the same instrumental configuration was attempted for collection of spectra in calibration and validation set, however, slight spectral differences occurred in both NIR and Raman spectra. If the possible spectral variation among spectral collections is admitted, the proper selection of a spectroscopic method capable of providing the higher selectivity and specificity should be a safer choice to guarantee the robustness of measurement, such as Raman spectroscopy, in this study.

4. Conclusion

With the use of more selective Raman features of components in a MAE solution, the corresponding concentrations

were more accurately predicted, although some minor spectral variations were present in the validation dataset. The use of selective Raman bands aided the prediction performance to be more robust and less vulnerable to spectral variation originating from the difference in instrumental configurations. Although NIR spectroscopy (direct spectral collection through a Teflon tube) had been utilized for monitoring the components in diverse etchant solutions, the issue of long-term prediction robustness should be critically addressed. Raman spectroscopy would be the preferred choice for on-line monitoring of etchant solutions as many etchant solutions contain hydrogen peroxide and inorganic acids that have distinct Raman features and Raman spectra can be also be directly collected from a Teflon tube as in NIR measurement. Future research will be directed to improve the robustness of Raman measurements by evaluating several subjects, such as the evaluation of the Teflon intensity variation resulting from the positional change of the Teflon tube and calibration transferability to other wet-bench systems using different Teflon tubes.

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